AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2018

Computer Science Paper II

Time: 2 hours 25 minutes Marks: 55

Constructed Response Questions (CRQs)

Question 1a:

Amjad has recently joined an organisation as a civil engineer. He needs a number of input and output devices to perform various tasks at the office. Suggest ONE appropriate device to perform each task.

i. He needs a device with built-in CAD programs for designers. The device would move pens of different colours in vertical and horizontal directions to draw on paper while the paper remains stationary.
ii. He needs a device that prints coloured printouts of documents containing photos with vivid colours and high contrast.
iii. He requires a device which converts handwritten documents into digital form which can be stored and viewed on the computer.

Question 1b:

Describe the following two phases of the system development life cycle.

i. System Analysis
ii. System Design

Question 2a:

Name a storage device that uses the sequential memory access method.

Question 2b:

Show whether the given statements are TRUE or FALSE. Justify your answer with the help of working.

(**Note:** No marks will be awarded if working is not shown)

Statement	TRUE/ FALSE	Working
1536 GB is greater than 1.5 TB.		
700 KB is smaller than 1 MB.		
56320 bytes is equal to 55 KB.		
5200 bits is greater than 650 bytes.		

Question 3a:

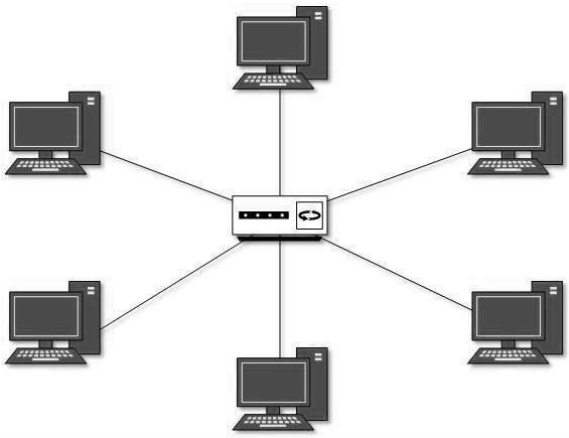
In the open system interconnection (OSI) model, the protocol data unit (PDU) is the data which moves from one layer of the OSI model to another. At different layers of the OSI model, the PDU is given different names.

Some OSI model layers and names of PDUs are given below. Match the appropriate layer with the corresponding PDU with the help of an arrow.

Name of OSI Model Layer	**Name of PDU**
Physical Layer	Segment
Data Link Layer	Packet
Network Layer	Bit
Transport Layer	Frame

Question 3b:

i. Name the device to which all the computers are connected in the given star topology.



ii. Mention any ONE advantage of using the bus topology instead of the star topology.

Question 4a:

Describe any THREE differences between peer-to-peer and client/ server network model.

Peer-to-Peer Model	Client/ Server Model

Question 4b:

The computer is an essential hardware component of a computer network.

List the names of any TWO other hardware components, other than the computer, that are required for a computer network.

Question 5a:

Write ONE difference between guided and unguided media and give an example of unguided media.

Question 5b:

Describe simplex and half-duplex communication mode and give an example for each.

Question 6:

Computer-controlled robots have minimised human labour in different industries.

Describe any FIVE advantages of using robots instead of humans in industries.

Question 7:

Describe any FIVE benefits of using blogs for teaching and learning.

Question 8:

i. Define the term 'search engine'.

ii. Zain wants to search for latest books on Computer Science (not older than one month) over the internet. The steps to carry out this search are given below. Arrange these steps in the correct sequence.

Step Number	Steps to Search a Book Using Internet	Correct Sequence Number
1	Type relevant keywords or phrases that best describe the book to be searched.	
2	Go through the list of fetched web pages to find the required content.	
3	Open any famous search engine in a web browser.	
4	Select book option and click the Search button to search the required book on the Internet.	

Question 9:

Consider the MS Excel table shown below.

	A	B	C	D	E	F	G
1	**Student Exam Performance**						
2	**Name of Student**	**Physics**	**Math**	**Computer Science**	**Maximum Marks**	**Obtained Marks**	**Percentage**
3	Kashif Ahmed	91	92	97	300	280	93.333333
4	Abdullah Tayab	76	62	55	300	193	64.333333
5	Sana Fatima	85	74	79	300	238	79.333333
6	Farah Khan	90	79	81	300	250	83.333333
7	Syed Jamal	51	59	61	300	171	57

By using the above-given table, write the MS Excel formula to

i. calculate Obtained Marks of Sana Fatima.
ii. calculate the average of obtained marks.
(**Note:** Do not use SUM or Average functions.)
iii. calculate the Percentage of Abdullah Tayab.
Mathematical formula of Percentage = (Obtained Marks ÷ Total Marks) ×100
iv. count the number of students who have obtained more than 70 marks in Math.

Extended Response Questions (ERQs)

The following questions offered a choice between part **a** and **b**.

Question 10a:

i. What is meant by ‘computer viruses’ and ‘computer safety’?
ii. Name any THREE anti-virus software.
iii. Describe FIVE measures through which we can ensure computer security.

Question 10b:

List FIVE different registers of a CPU. Describe the tasks performed by EACH to improve the performance of the CPU.